

AI/ML BOOTCAMP

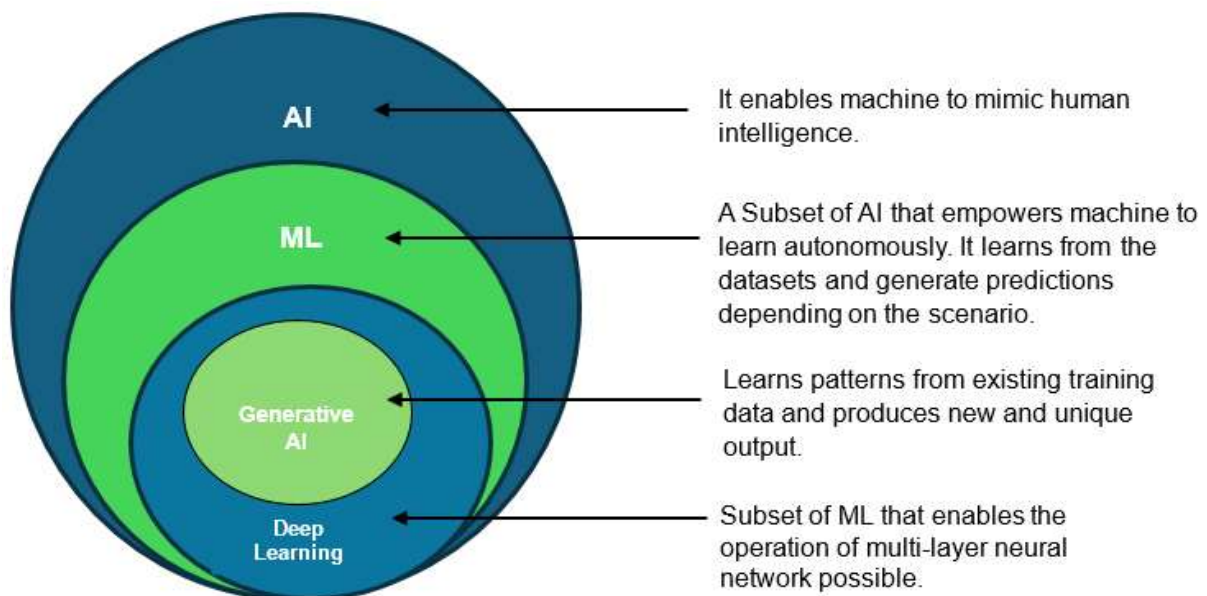
(Day -1)

1st concept: AI VS ML VS DEEP LEARNING:

AI: AI refers to the intelligence model which mimics humans and tries to make decision based in the learning.

ML: It refers to the knowledge gained by model through continuous learning and understanding.

Deep learning: It works as the brain of AI which helps AI to use its knowledge and to implement knowledge by recognising patterns.



2nd concept: what are Deep fakes?

Deep fakes-manipulated videos or images that convincingly alter the likeness or voice of a person—have become a phenomenon in both popular culture and the broader landscape of digital media. What once seemed like a distant concept found only in the realm of science fiction has now become a powerful, and at times terrifying, tool with wide-reaching consequences.

Deep fakes are media (typically videos, but also images or audio) that have been altered through the use of deep learning algorithms to manipulate or fabricate visual and auditory content in a hyper-realistic manner. The term

“deepfake” is a portmanteau of “deep learning” and “fake,” referring to the AI-driven processes that create these altered media.

The Ethical Concerns Surrounding Deep fakes:

- *Privacy and consent*
- *Misinformation*
- *Impact on trust on media*
- *Violating ethics in the field of creativity.*
- *Exploitation and harm.*

Deep Fake technology is a specific application of AI that involves creating manipulated or fabricated content, usually in the form of images, videos, or audio recordings. Deep Fake algorithms use machine learning techniques to generate highly realistic simulations that can deceive viewers into believing that the manipulated content is real. These simulations can mimic someone's appearance, voice, or actions, leading to potential misuse and exploitation.

Difference between GenAI and Deep fake:

Generative AI (Gen AI): This is an umbrella term for any artificial intelligence that can create *new* content from scratch based on a prompt. It includes text (ChatGPT), code, music, 3D models, and artwork (Midjourney).

Deep fakes: This is a narrow, specialized application of Gen AI. It specifically refers to synthetic media (videos, images, or audio) where a real person's likeness, face, or voice is replaced or altered to make them say or do something they never actually did.

Generative AI: The goal is **creation and productivity**. It's used to write essays, design video game assets, generate marketing copy, or automate repetitive tasks. It is generally built to be a helpful tool.

Deep fakes: The goal is almost always **imitation and representation**. While they can be used for fun (like movie special effects or memes), they are frequently used to mislead, spread disinformation, commit fraud, or impersonate someone without their consent.

3rd Concept: AI CYCLE –

AI cycle comprises of:

- *Problem Scoping – **Data engineer***
- *Data collection - **Data engineer***

➤ <i>Big data – Big data engineer</i>
➤ <i>Data – data engineer</i>
- *Data Analysis – **Data Analyst | Power BI Analyst***
- *Data modelling – **Data scientist***
- *Data evaluation – **Data scientist***
- *Deployment- **ML engineer***